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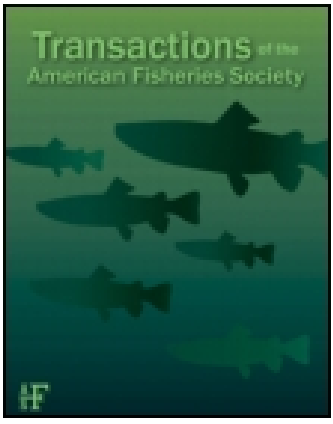

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Direct and Indirect Effects of Predation on a Fish Community: A Whole-Lake Experiment

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Direct and Indirect Effects of Predation on a Fish Community: A Whole-Lake Experiment

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Abstract.—We designed an experiment to test the relative importances of direct and indirect effects of piscivorous predation on an assemblage of small fishes in a piscivore-free lake. The direct effect is defined as consumption of prey fishes. Indirect effects include habitat changes associated with predator avoidance behavior, increases of emigration rates, and changes of composition and size structure of the prey community. After a year of premanipulation study, adult northern pike *Esox lucius* (a predator species) were introduced into the lake. Decrease of prey fish biomass due to increased emigration was at least as great as that due to direct consumption by the predators. Significant indirect effects included decreased abundance of dominant species, increases of some rare species, and decreases of mean size for species most vulnerable to predation. Indirect effects occurred rapidly and were most apparent during weeks immediately after northern pike were introduced.

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The direct effect of piscivores on community composition and the population dynamics of prey fishes is widely documented in a growing literature based primarily on the behavioral and evolutionary ecology of predator–prey systems (e.g., Clepper 1979; Zaret 1979; Werner et al. 1983; Power et al. 1985; Sih et al. 1988). Indirect (nonlethal) effects have received only limited attention but may be equally if not more important at the community and ecosystem levels (Popova 1978; McQueen et al. 1986; Carpenter et al. 1987; Mills et al. 1987).

In open systems, a new predator can evoke changes in habitat selection by prey, thereby altering the effective community composition and food web structure. For example, strong predator-avoidance behavior was observed after a substantial density of planktivorous minnows (about 20,000/hectare) was introduced into a lake with only a small population of piscivores (less than 30/hectare; Carpenter et al. 1987). The minnow population declined much more rapidly than could be accounted for by the direct consumption by known numbers of predators (He 1986). The introduced minnows had an immediate effect on zooplankton prey, but it was brief because the minnows abandoned the pelagic zone, retreated to littoral refugia, and, within less than a month, were functionally insignificant as zooplanktivores (Carpenter et al. 1987).

Nocturnal migrations from the littoral to pelagic zones have been a predator-avoidance response by prey fishes in large lakes (Goodyear 1973; Stuntz 1975). Predation risk was largely responsible for the habitat shifts of juvenile fishes observed in experimental ponds (Werner et al. 1983). In stream systems, predatory fish can alter the behavior and distribution of herbivorous fish to the extent that filamentous algal growth indicates the presence of the piscivore in individual pools (Power et al. 1985; Schlosser 1987). In each of the above examples, the direct effects of prey fishes on their primary food resources were strongly modified by the indirect effects of a piscivorous predator. Such changes can occur at any trophic level below the top one (Mittelbach and Chesson 1987; Mittelbach 1988).

In lentic systems, fish community structure is determined both by abiotic factors and by species interactions that can cause local extinctions, prevent successful invasions, or both. Predation is particularly important in this regard (Murdoch and Bence 1987). In the lakes of the upper midwestern USA, surveys and comparative studies have revealed that fish community structure is strongly related to the presence of a self-sustaining piscivore population (Tonn and Magnuson 1982; Rahel 1984). Lakes without a piscivore are dominated by dense populations of typically small fishes whose life histories, morphologies, and behaviors make them superior competitors. Larger, longer-lived fishes (including many with well-developed antipredatory morphologies) may persist in these habitats, but usually at low densities (Tonn and Magnuson 1982). A species-by-species experimental analysis of predatory or competitive interactions in these diverse communities is neither practical nor appropriate, because it removes in-

teractions from the appropriate community and from spatiotemporal scales in which other important mechanisms may operate (Frost et al. 1988).

We argue that the effects of piscivores on the prey fish community should be evaluated from two perspectives: (1) direct effects due to consumption of prey, and (2) indirect effects such as behavioral responses of prey fishes to the presence of piscivores, including migrations, changes of foraging or other activity patterns, changes in size structure of prey species, and community changes that integrate such responses (Stein 1979; Helfman 1986).

It is usually difficult to evaluate the relative importances of direct and indirect effects, because studies must be conducted on the scale of whole systems, and thus require substantial detail on population dynamics of the predator, its primary prey species, and the effects of strong interactions imposed by other members of the community on the predator or prey species (Paine 1980). In addition, natural experiments rarely provide clear evidence because many other factors such as diffuse competition, variable recruitment, lags in numerical responses, or abiotic disturbances can affect results. Whole-lake manipulations seem to be an effective way to reduce these problems (Zaret 1979; Helfman 1986; Carpenter and Kitchell 1988; Kitchell et al. 1988).

In this paper, we report the outcome of a whole-lake experiment on the direct and indirect effects of an introduced predator (northern pike *Esox lucius*) on a diverse fish community. We focused on species-specific and behavioral responses of prey fish populations as a means of assessing effects on community structure. Using the statistical techniques of randomized intervention analysis (Carpenter et al. 1989), we conducted a quantitative test of the hypothesis that direct and indirect effects of predation are equally important.

Study Lake

The study site, Bolger Bog, is a small lake located within the University of Notre Dame Environmental Research Center near Land O'Lakes, Wisconsin. The lake is surrounded by bog mat and has an open water area of about 1 hectare. It is shallow (maximum depth, 4.0 m), acidic (pH 6.0), and has low transparency (secchi depth, 0.25–0.7 m). The thermocline has a depth of 0.5–1.0 m and is well developed between May and late September.

One small inflow channel extends about 200 m from the lake and gathers groundwater. The outflow channel (1–2 m wide) connects to a nearby stream. Fish can migrate between the lake and the stream through the outflow channel. However, the channel is a potentially severe habitat for fish because oxygen concentrations may decline to less than 1 mg/L at night (He, unpublished data).

The fish community is diverse, consisting mainly of small planktivores and omnivores (Table 1), and is typical of the *Umbra*-cyprinid assemblage in lakes of this region (Tonn and Magnuson 1982). Fish communities similar to that in Bolger Bog are common in lakes that are subject to regular winterkill and, as a consequence, lack a significant piscivore population. To avoid winter hypoxia, most fish species emigrated from the lake in late autumn and early winter, then returned in spring (Magnuson et al. 1985). No large piscivorous fishes are present (except when stocked experimentally), although some predation undoubtedly occurs due to local populations of piscivorous birds and mammals. The zooplankton community is dominated by predatory (e.g., *Asplanchna* spp.) and other rotifers, *Bosmina longirostris*, small copepods, and *Chaoborus* spp. *Daphnia* spp. are rare in this lake (S. R. Carpenter, University of Notre Dame, and Kitchell, unpublished data).

Methods

Experimental design and sampling.—The experiment was conducted from late May to late October in 1986 and 1987. The first year served as a baseline for experimental manipulations during the second year. Limnological data, including oxygen, temperature, chlorophyll *a*, and secchi-disk depth, were sampled weekly from late May to late August at a fixed central station in the lake. Prey fish were sampled with minnow traps (height, 42 cm; diameter, 22.5 cm; mesh, 6 mm). Each sample included catches from 6 traps set in each channel, 12 traps set along the perimeter of the lake, and 9 traps set on transect lines in the open water. All traps were set just below the surface for 22–24 h. All fish caught in each trap were identified to species, counted, and released. Samples were taken every 2–3 weeks in 1986 and weekly in 1987. In 1987, each sample was replicated the following day, and averages of the two samples were used in the later analysis. The relative prey fish densities were estimated from catches per unit effort (i.e., numbers of fish caught per trap-hour) with minnow traps. In 1986, about 30 fish of each species were taken for length and weight measurements on each sampling date. Seasonal mean weights for each species were then calculated.

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TABLE 1.—Fish species caught by minnow traps in Bolger Bog, Michigan, 1986–1987.

Common name	Scientific name	Length range (mm)
Northern redbelly dace	*Phoxinus eos*	45–85
Central mudminnow	*Umbra limi*	50–120
Fathead minnow	*Pimephales promelas*	50–90
Brook stickleback	*Culaea inconstans*	45–75
Finescale dace	*Phoxinus neogaeus*	42–90
Brassy minnow	*Hybognathus hankinsoni*	40–80
Golden shiner	*Notemigonus crysoleucas*	70–140[a]
Blacknose shiner[b]	*Notropis heterolepis*	50–75[a]
Bluegill	*Lepomis macrochirus*	30–60
Pumpkinseed[c]	*Lepomis gibbosus*	30–60[a]
Creek chub	*Semotilus atromaculatus*	70–140[a]
Iowa darter[b]	*Etheostoma exile*	50–70[a]
Pearl dace[b]	*Semotilus margarita*	70–140[a]

[a] Estimated.
[b] Very rare.
[c] Pooled with bluegill.

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As a test of predator effects on the size composition of prey species, we recorded length frequencies for redbelly dace *Phoxinus eos*, the most abundant species. Total lengths (nearest millimeter) were measured for up to 200 individuals randomly selected from catches in both the lake and the outflow channel on each sampling day in 1987. All fish were released after measurement. No length-frequency data were collected after 23 June 1987 because few redbelly dace were caught.

In June 1986, a mark-and-recapture method (Petersen method, Ricker 1975) was used to estimate the population density of redbelly dace. This estimate calibrated catch per unit effort with fish density in subsequent sampling. Densities of other species were estimated by comparison of their catches per unit effort to those of redbelly dace. Total prey fish biomass was then estimated from the density and mean weight of each species.

On 23 May 1987, 15 adult northern pike (mean length = 507 mm, SD = 30 mm) were removed from nearby Morris Lake by hook and line and introduced to Bolger Bog. Each northern pike was measured, weighed, and tagged before release. The outflow channels were blocked by 2.5-cm-mesh wire fence. The fences kept northern pike in the lake but allowed prey fishes to get through. Northern pike diets and growth rates were determined on 29 June 1987 and again on 29 July 1987; an overnight gill-net set was used to capture nine individuals on each date. We replaced these individuals with fish of similar sizes. The allometric condition factor, $K_n = W/L^b$ (Ricker 1975)—W = weight, L = length, b = a population-specific exponent—was compared for northern pike caught the same day from both Bolger Bog and Morris Lake (t-test of $\log_{10} K_n$). To test the recolonization rate of minnows, we used gill nets to remove all northern pike from the lake in late August 1987.

Data analysis.—Length frequencies from the channel and open-lake habitats were compared by χ^2 tests to determine if the presence of northern pike in the lake resulted in changes to the size structure of the redbelly dace population.

The catches per unit effort of individual species and of all species combined were compared between the two years. Standard errors of catch per unit effort for each sampling date were computed from all traps set at the same date. Ratios of catch per unit effort in the outflow channel to that in the lake were used to determine habitat preference by prey species. Because the outflow channel was shallower and had 2–4 mg/L lower oxygen content than the lake, we expected that fish would prefer the lake to the channel habitat. Therefore, we interpreted higher catches of prey fishes in the channel to indicate predator-avoidance behavior by prey species that moved to or through the channel to escape the northern pike.

In the analysis, the lake was considered as the manipulated system and the outflow channel as the reference system. Premanipulation data were from the period before the addition of northern pike on 23 May 1987. Postmanipulation data were from the period when northern pike were in the lake (28 May–28 August 1987). The catch per unit effort for each and all prey species from the lake and channel were then subjected to randomized intervention analysis (Carpenter et al. 1989). This method detects changes in a manipulated system relative to a reference system after a manipulation. A time series of differences between systems is calculated, along with mean intersystem differences (D-values) before and after the manipulation. The absolute value of the difference between premanipulation and postmanipulation D-values is the test statistic. This method does not require replication in the experiment, because no distribution assumption is made for the test statistics. Because the intersystem mean is used in this method, year-to-year variability is removed.

Estimate of indirect effects.—We used the emigration of prey fishes to represent indirect effects quantitatively. The cumulative biomass change due to emigration at time t, $Be(t)$, was

$$Be(t) = BT(t) - [Bp(t) + Bc(t)];$$

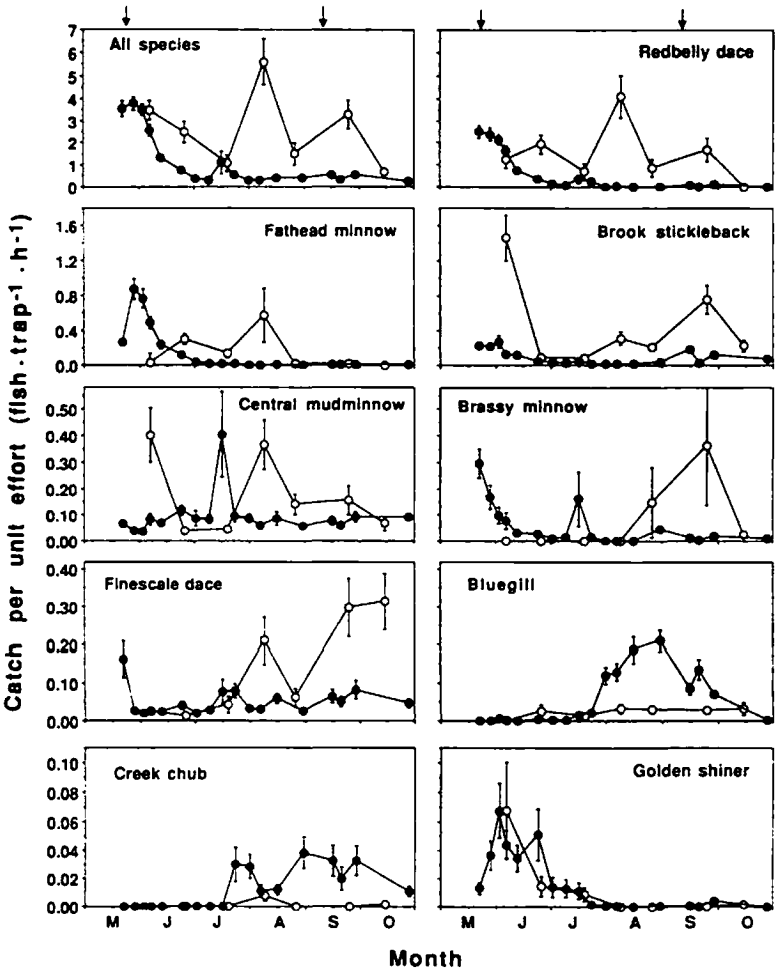

FIGURE 1.—Catch per unit effort of nine fish species and all species combined for Bolger Bog in 1986 (open circles) and 1987 (solid circles). Vertical bars show standard errors for each sampling date. Dates of addition and removal of northern pike in 1987 are indicated on top panels by arrows.

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$BT(t)$ = estimated total prey biomass at time t without northern pike predation;
$Bp(t)$ = prey biomass in the lake at time t;
$Bc(t)$ = cumulative prey biomass consumed by all northern pike until time t.

Expressed in words, the equation can be written

$$\begin{matrix}\text{Biomass}\\ \text{change}\\ \text{due to}\\ \text{emigration}\end{matrix} = \begin{matrix}\text{Biomass}\\ \text{without}\\ \text{predation}\end{matrix} - \left[\begin{matrix}\text{Biomass}\\ \text{with}\\ \text{predation}\end{matrix} + \begin{matrix}\text{Biomass}\\ \text{consumed}\\ \text{by pike}\end{matrix}\right].$$

To estimate $BT(t)$, one must know the natural mortality rate, the recruitment rate to trappable size (>45 mm), and the emigration and immigration rates of species in the absence of predation by northern pike. To estimate these rates for 1987, we used the biomass values of the same period in 1986 as an approximation, because the total prey biomass in the beginning of the summer was similar in 1986 and 1987 (Figure 1). We estimated $Bp(t)$ using the catch per unit at time t. $Bc(t)$ was estimated as

$$Bc(t) = [1 - M(T)] \sum_{p=1}^{N} \sum_{t=1}^{T} B(t, p);$$

$M(T)$ $[0 \leq M(t) \leq 1]$ is the mean natural mortality rate of prey biomass consumed by northern pike during the period $t = 1$ to $t = T$; $B(t,p)$ is biomass consumed by an individual northern pike (p) at time t (estimated from a bioenergetics model; Kitchell et al. 1977; Bevelhimer et al. 1985); T is the time duration over which cumulative consumption is calculated; and N is the total number

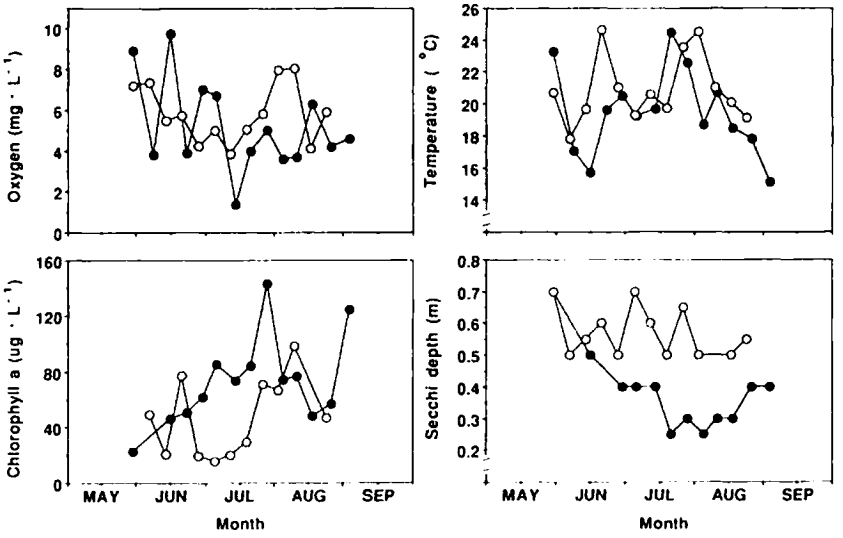

FIGURE 2.—Mean epilimnion oxygen concentration, temperature, chlorophyll *a*, and secchi-disk depth in Bolger Bog in 1986 (open circles) and 1987 (solid circles).

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of northern pike in the lake during time T. The term $[1 - M(T)]$ accounts for those of the prey eaten by northern pike that would have died of natural causes in the predator's absence. Because $M(T)$ is difficult to estimate, $Bc(t)$ was calculated for different values of $M(T)$.

The ratio $R = Be(t)/Bc(t)$ was then used to evaluate the relative importance of indirect and direct effects. That is,

$$R = \frac{\text{biomass change due to emigration}}{\text{biomass change due to consumption by pike}}.$$

If R was greater than 1, indirect effects were greater than the direct effect of predation. If R lay between 0 and 1, predation by northern pike was more important than emigration. Values for R of less than 0 result from negative $Be(t)$ values, which indicate that immigration may have occurred.

Results

Fish Densities

Introduction of northern pike in May 1987 significantly reduced total prey fish density (Figure 1). The annual mean catch per unit effort for all species in 1987 was reduced to 47% of that observed in 1986. No statistical analysis was used to test for differences in catch per unit effort between the two years, because the effects of northern pike may have been confounded with other year-to-year variability. However, we believe that the northern pike addition was the major cause for the reduction of catch per unit effort in 1987 because (1) catch data for all species combined and for the dominant species (redbelly dace) were similar at the beginning of both years (Figure 1), and (2) major limnological factors, such as oxygen concentration, temperature, and chlorophyll-*a* concentration, were also similar in both years (Figure 2). Although secchi-disk depths during July and August of 1987 were greater than those for the same period of 1986, we believe that the difference resulted from observations made by different persons in the two years.

Changes in catches of prey fish in 1987 were highly species-specific. The greatest reductions occurred in four species: redbelly dace, fathead minnow, finescale dace, and brassy minnow (Figure 1). The early-June density of redbelly dace was higher in 1987 than in 1986 and decreased after the northern pike addition. Only 1 month after the introduction, redbelly dace were rarely found in the lake, and their density remained very low throughout the summer. Fathead and brassy minnows showed similar responses. Density of finescale dace decreased immediately after the addition in 1987 and remained low throughout the summer. Finescale dace density increased after August 1986, but a similar increase did not occur in 1987. Brassy minnow density was much higher in the beginning of 1987 than in 1986 and declined after the addition of northern pike (Figure 1).

Golden shiners did not show differences be-

tween the two years. The densities of central mudminnows and brook sticklebacks were lower in early June 1987 than in 1986, and did not show an increasing trend after August as they did in 1986. Densities of creek chubs and bluegills were higher in 1987 than in 1986. Catches of these two species increased in late July 1987 (Figure 1).

Community Composition

In 1986, the fish community was dominated by redbelly dace on all but the last sampling date (Figure 3). The mean percentage of redbelly dace in the 1986 catch was 60%, excluding the last sampling date (11 October). Catches of redbelly dace also declined in late October 1987, after they had slightly increased in September. We believe that redbelly dace begin to migrate out of the lake in October to avoid winter hypoxia (Magnuson et al. 1985). In late May and early June 1987, redbelly dace made up about 65% of the community, as they had in June 1986. As the catch of prey fishes declined into late July 1987, redbelly dace dropped to a small fraction (2–20%) of the remaining assemblage. Catches of bluegills and creek chub together increased during 1987 to a high of 64% of the total catch by late August. After northern pike were removed from the lake in late August, combined fractions of fathead minnow, finescale dace, and brassy minnow increased to 26%, and combined fractions of central mudminnow, brook sticklebacks, and golden shiners increased to 40%. Catches of redbelly dace increased to 19%. During September, the prey fish community had the highest diversity and was no longer dominated by any single species.

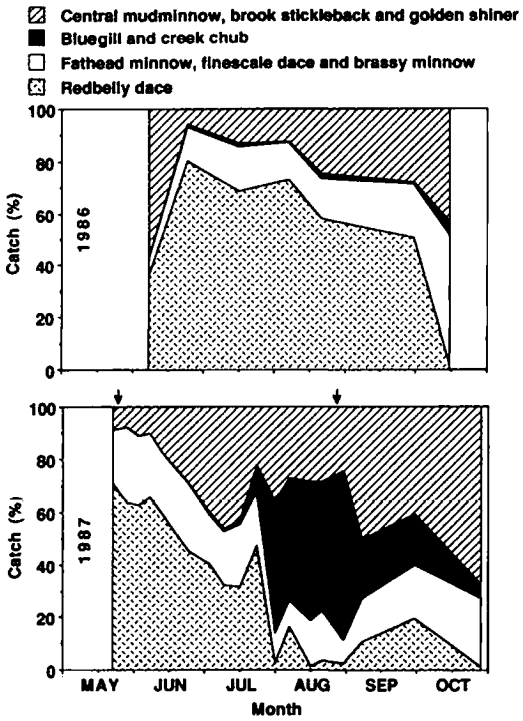

FIGURE 3.—Species composition, as percentages of total catch, for four species groups in Bolger Bog in 1986 and 1987. Species were grouped according to similarity of catch per unit effort in both years. Dates of addition and removal of northern pike in 1987 are indicated by arrows on the top of lower panel.

Behavioral Responses

More fish moved to the outflow channel after the northern pike were added in late May 1987, as indicated by higher ratios of catches in the outflow channel to those along the lake perimeter (Figure 4). Beginning in June 1987, catch ratios increased to a maximum of 8.6 before decreasing in July. In the same period in 1986, the ratios decreased to a low of 0.17. These behavioral responses differed among species. Four species showed significant shifts between habitats (Figure 5). More redbelly dace, finescale dace, and brassy minnows shifted from the lake to the outflow channel than stayed in the lake after the addition of northern pike. More bluegills shifted from the outflow channel to the lake than went the other way. Other species showed no significant movement into either habitat (Figure 5).

Effects on Size Structure of Redbelly Dace

Comparison of redbelly dace length frequencies between the outflow channel and the lake indi-

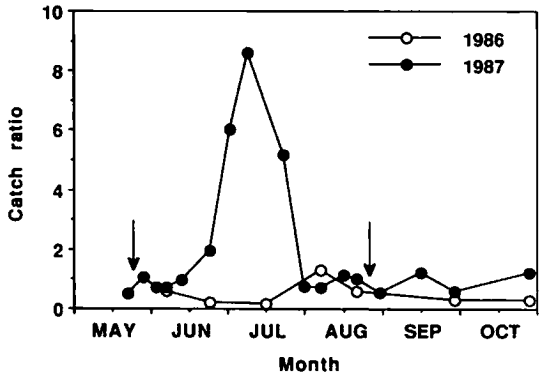

FIGURE 4. Ratios of catch per unit effort (number of fish/trap-hour) for minnow traps set in the outflow channel to that of traps set along the perimeter of Bolger Bog. Data are for all species. Dates of addition and removal of northern pike in 1987 are indicated by arrows.

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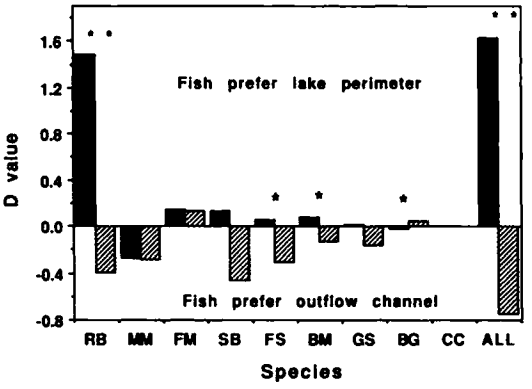

FIGURE 5.—Results of randomized intervention analysis of differences in catch per unit effort (number of fish/trap-hour) between the outflow channel and the perimeter of Bolger Bog. The D values are differences between means for lake perimeter and outflow channel before (black) and after (crosshatched) addition of northern pike. Positive D values indicate that fish prefer the lake perimeter; negative D values indicate that fish prefer the outflow channel. Asterisks indicate significant values ($0.05 \geq P^{*} > 0.0001$; $P^{**} \leq 0.0001$). Species abbreviations are RB = redbelly dace, MM = central mudminnow, FM = fathead minnow, SB = brook stickleback, FS = finescale dace, BM = brassy minnow, GS = golden shiner, BG = bluegill, CC = creek chub, and ALL = all species combined.

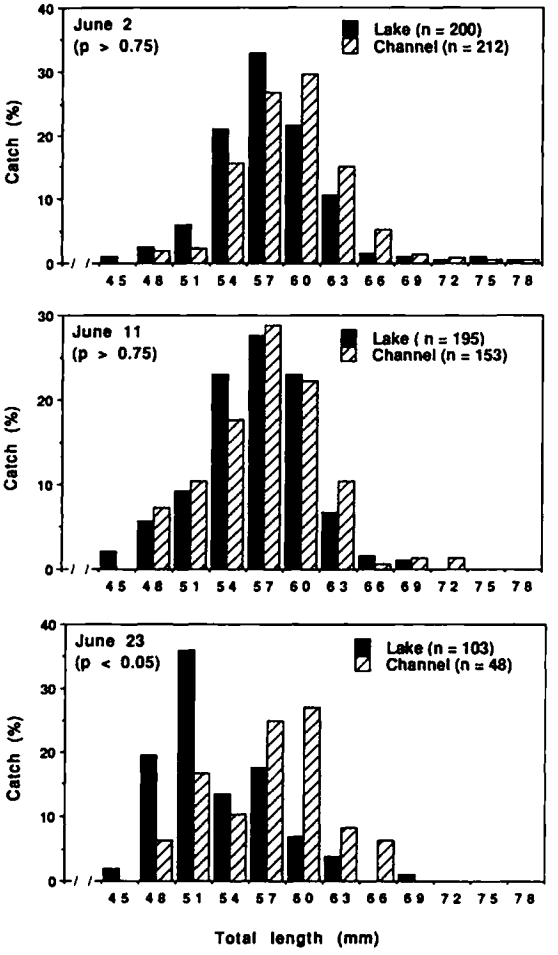

FIGURE 6.—Length-frequency distributions for redbelly dace, presented as catch percentages, from lake perimeter and outflow channel habitats of Bolger Bog for three days in June 1987. Fish were grouped in 3-mm size intervals. Numbers of fish measured (n) are given for each sample; p values are from χ^2 tests.

cated that the northern pike addition had size-specific effects. On 2 and 11 June 1987, length-frequency distributions from the lake and outflow channel were not significantly different (χ^2 test, $P > 0.75$), although the mean lengths of redbelly dace from the outflow channel were slightly greater than those from the lake (Figure 6). On 23 June, about 4 weeks after the addition of northern pike, the mean length of redbelly dace in channel traps was significantly greater than that of lake fish (t-test of mean, $P < 0.05$), and the length-frequency distributions were significantly different (χ^2 test, $P < 0.05$).

Growth of Northern Pike

The average growth rates of northern pike were 6.95–8.63 g/d from late May to late August (Table 2). On 28 June and 29 July, northern pike of similar length from Bolger Bog were much heavier than those from Morris Lake. The allometric condition factors, K_n, of northern pike from Bolger Bog were significantly greater than those for northern pike from Morris Lake ($P < 0.001$). Examination of stomach contents of northern pike provided little information on prey preference, because 45% of northern pike caught in gill nets had empty stomachs. Although our minnow traps caught only prey fishes in the size range of 45–125 mm, all identifiable prey items in the stomachs of northern pike were in the length range of 55–100 mm. We assume that northern pike consumed negligible amounts of prey fish smaller than 45 mm.

Direct versus Indirect Effects

Ratios of biomass loss due to emigration to biomass consumed by northern pike revealed that indirect effects of predation were much greater than direct effects ($R > 1$) in June for all $M(t)$ values tested (Figure 7). The R-value was greatest 2 weeks after northern pike were introduced and then

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TABLE 2.—Growth of northern pike in Bolger Bog in 1987.

Period	Weight range (g)		Mean growth, g/d (SD)	N
	Start of period	End of period		
23 May–29 Jun	500–1,160	780–1,300	8.63 (2.68)	9
29 Jun–29 Jul	500–1,020	740–990	6.95 (1.19)	9
29 Jul–21 Aug	510–600	710–870	7.79 (1.38)	4[a]

[a] Does not include fish caught by gill nets but partially consumed by turtles or otters.

gradually decreased. During the period when R exceeded 1, fish biomass in the lake declined to 11% of the maximum biomass, which occurred in late May (Figure 1). Direct effects were greater than indirect effects ($0 < R < 1$) from early July or $M(t)$ less than or equal to 0.2, from mid-July for $M(t)$ less than or equal to 0.5, and from early August for $M(t)$ less than 0.8 (Figure 7). Direct effects were greater than indirect effects only after total community biomass was low (<11% of initial biomass).

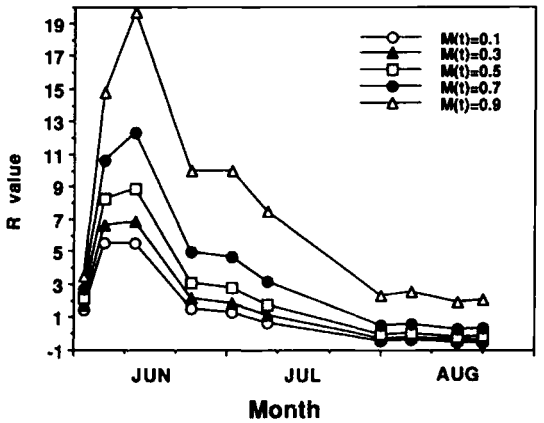

FIGURE 7.—Values of R for different mortality rates, $M(t)$, for fish in Bolger Bog in 1987; R is the ratio of biomass change due to emigration over biomass consumed by northern pike. Ratios greater than one indicate greater indirect effects than direct effects.

Discussion

The density of northern pike introduced into Bolger Bog (15/hectare) might have been higher than that of most lakes subject to public sport fishing, but it was not unusual for unexploited natural lakes. For example, northern pike density in Morris Lake (area, 5.7 hectares) was greater than 25/hectare during 1986–1988 (our personal observation). Density of northern pike can reach 7.3/hectare in Big Cedar Lake (406 hectares) in east-central Wisconsin (Priegel and Krohn 1975). In lakes where northern pike are stocked, densities can be as high as 115/hectare (Murphy Flowage, Wisconsin, 72 hectares; Snow 1974), or as low as 5.7/hectare, as in Horseshoe Lake, Minnesota (344 hectares), where northern pike were about 37% of the total piscivore community that included walleye *Stizostedion vitreum* and largemouth bass *Micropterus salmoides* (Anderson and Schupp 1986).

Our results suggest that redbelly dace, fathead minnow, finescale dace, and brassy minnow are more vulnerable to predation by northern pike than bluegill, brook stickleback, creek chub, and golden shiner. This may be explained by differences in prey size and morphology. Bluegills and brook sticklebacks are spiny-rayed fishes. Northern pike usually prefer soft-rayed prey and select minnows (fathead minnows, goldfish *Carassius auratus*, common carp *Cyprinus carpio*) over bluegills and yellow perch *Perca flavescens* (Beyerle and Williams 1968; Mauck and Coble 1971; Coble 1973). Tiger muskellunge (F_1 hybrid of male northern pike and female muskellunge, *Esox masquinongy*) require four times as many strikes and longer pursuits to capture bluegills than to catch fathead minnows (Moody et al. 1983). Golden shiners and creek chub are larger than other prey in Bolger Bog (mean length > 100 mm) and may escape more efficiently from northern pike. In laboratory experiments, northern pike could not be conditioned to eat golden shiners or bluegills when other prey (gizzard shad *Dorosoma cepedianum*, common carp, and fathead minnows) were present (Mauck and Coble 1971).

Many laboratory experiments have shown that behavioral responses to a predator may develop rapidly (Fraser and Cerri 1982; Moody et al. 1983; Savino and Stein 1989). Some prey fishes move to a refuge area or display anti-predator behavior within minutes to hours after predators are introduced (Magurran and Pitcher 1987; Sih et al. 1988; Savino and Stein 1989). Our experiment demonstrated that behavioral phenomena that have been observed over short time scales (minutes, hours, days), over small spatial scales (aquarium, laboratory stream, experimental pond), and in simple systems (few prey) may also occur over the long term (weeks to months), and in whole lakes with a more biologically complex system (multiple prey species). In our study, responses at the scale of a whole lake were as dynamic as those in experimental apparatus, although they took somewhat longer, lending encouragement to the pros-

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pect of applying lessons from small-scale experimental studies to predictions at larger scales.

In fisheries management, stocking has been used extensively to establish and sustain piscivore populations (Snow 1974; Priegel and Krohn 1975; Anderson and Schupp 1986). Studies of prey community responses to predators are important to understanding the dynamics of stocked piscivores and their effects on prey fishes (Anderson and Schupp 1986). Traditional studies usually pay more attention to changes in prey fish density than to behavioral responses of prey to predation. Our study suggests that behavioral responses are important factors in determining the decline of prey fish populations. Although behavioral responses may be dramatic only in the short term, they happen when introduced predators are adapting to a new environment and, therefore, may be a key factor in determining the outcome of introductions in complex and diverse communities.

A survey of northern Wisconsin bog lakes showed that golden shiner was the only cyprinid species that coexisted with centrarchid predators. Redbelly dace, finescale dace, fathead minnow, blacknose shiner, and brook stickleback were not found in lakes that had northern pike or centrarchid predators (Rahel 1984). Thus, species-selective predation was probably the major force in changes of the fish community in Bolger Bog in 1987. Composition of fish communities in other lakes of the region reinforces our interpretation of the importance of predation in determining fish community structure.

In summary, our experiment demonstrated that the addition of a predator not only causes declines in prey fish abundance, it also changes community structure. Emigration of prey fishes accounted for both a rapid decline of total fish density and a change in community composition. Perhaps more importantly, indirect effects due to prey behavioral responses were much more pronounced than the direct effects due to predation, because consumption by northern pike remains relatively constant over the summer (Kitchell et al. 1977; Bevelhimer et al. 1985). Indirect effects must be considered major responses when the short-term effects of predation are evaluated, and as unseen components that prevent or inhibit development of species that might dominate in the absence of the predator. This "keystone" effect derives from strong interactions (Paine 1980) that persist to the point of local extinctions and have important effects on ecosystem processes (Carpenter and Kitchell 1988; Mazumder et al. 1990).

If northern pike had persisted and reproduced in Bolger Bog, the fish community of the lake now would be fundamentally different from that observed in 1986. Extension of that logic to other trophic levels requires a better understanding of the compensatory response capacity of the entire community. Understanding and predicting those responses remain among the most substantial challenges for future work.

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